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(54) **INSERTION DEVICE POSITIONING
GUIDANCE SYSTEM AND METHOD**

(71) Applicant: **ENVUE MEDICAL HOLDINGS,
CORP.**, Arlington Heights, IL (US)

(72) Inventor: **Doron Besser**, Tel Aviv (IL)

(73) Assignee: **ENVUE MEDICAL HOLDINGS,
CORP.**, Dover, DE (US)

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(Continued)

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CPC **A61B 34/20** (2016.02); **A61B 5/062**
(2013.01); **A61B 5/743** (2013.01);
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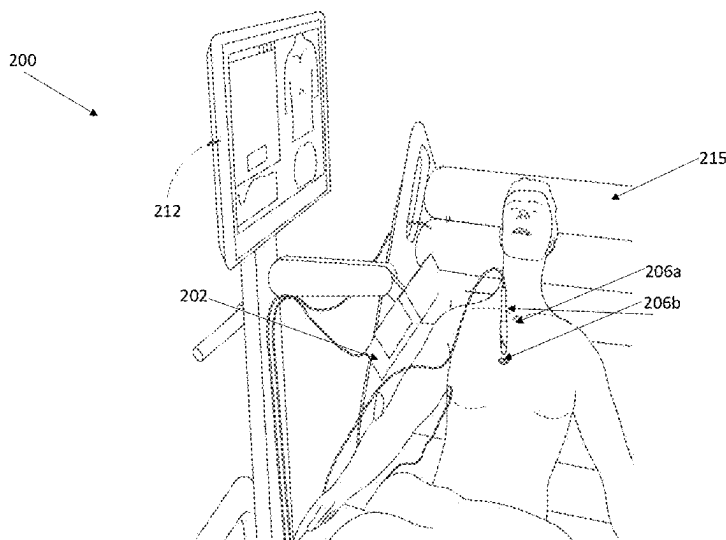
Primary Examiner — Mark D Remaly

(74) *Attorney, Agent, or Firm* — The Roy Gross Law
Firm, LLC; Roy D. Gross

(57) **ABSTRACT**

There is provided herein a guidance system for positioning an insertion device comprising: an electromagnetic field generator configured to generate an electromagnetic field covering a treatment area, an insertion device comprising an electromagnetic sensor, the electromagnetic sensor configured to receive signals indicative of the electromagnetic field, and a processing circuitry configured to: load an X-ray, CT, ultrasound or MRI image of the subject's chest, mark a location of a first and a second anatomic landmarks on the subject's torso using a registration sensor and obtaining a subject coordinate system based thereon, identify the location of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image of the subject's chest; aligning the subject coordinate system with the loaded X-ray, CT, ultrasound or MRI image, and display, on the image, a path of the insertion device insertion with respect to the first and the second anatomic locations; wherein the path is generated according to changes in the strength of the electromagnetic field sensed by the tip sensor's during the insertion of the insertion device.

15 Claims, 14 Drawing Sheets
(4 of 14 Drawing Sheet(s) Filed in Color)



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- continuation-in-part of application No. 17/328,642, filed on May 24, 2021, now Pat. No. 11,779,403, which is a continuation of application No. 16/561,320, filed on Sep. 5, 2019, now Pat. No. 11,045,260.
- (60) Provisional application No. 62/746,854, filed on Oct. 17, 2018.
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A61M 25/01 (2006.01)
A61B 1/273 (2006.01)
- (52) **U.S. Cl.**
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(58) **Field of Classification Search**

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See application file for complete search history.

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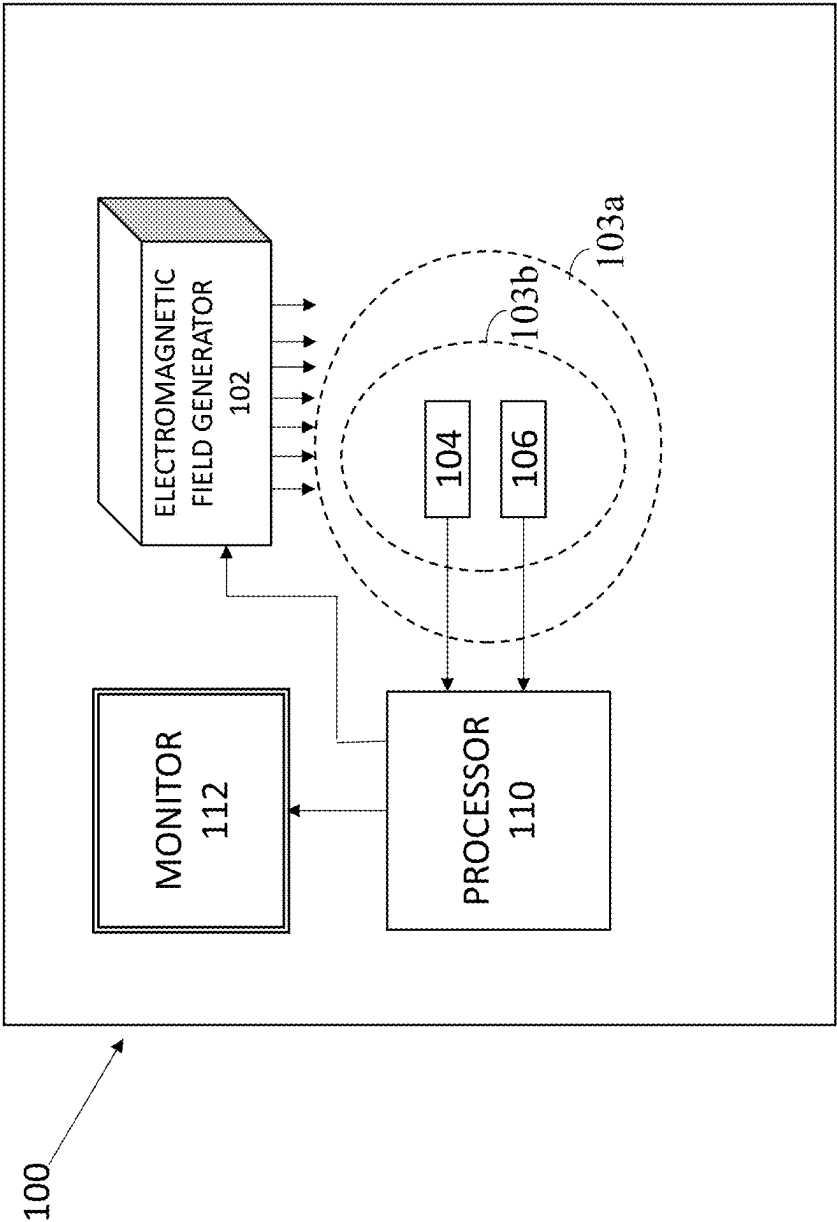


FIG. 1

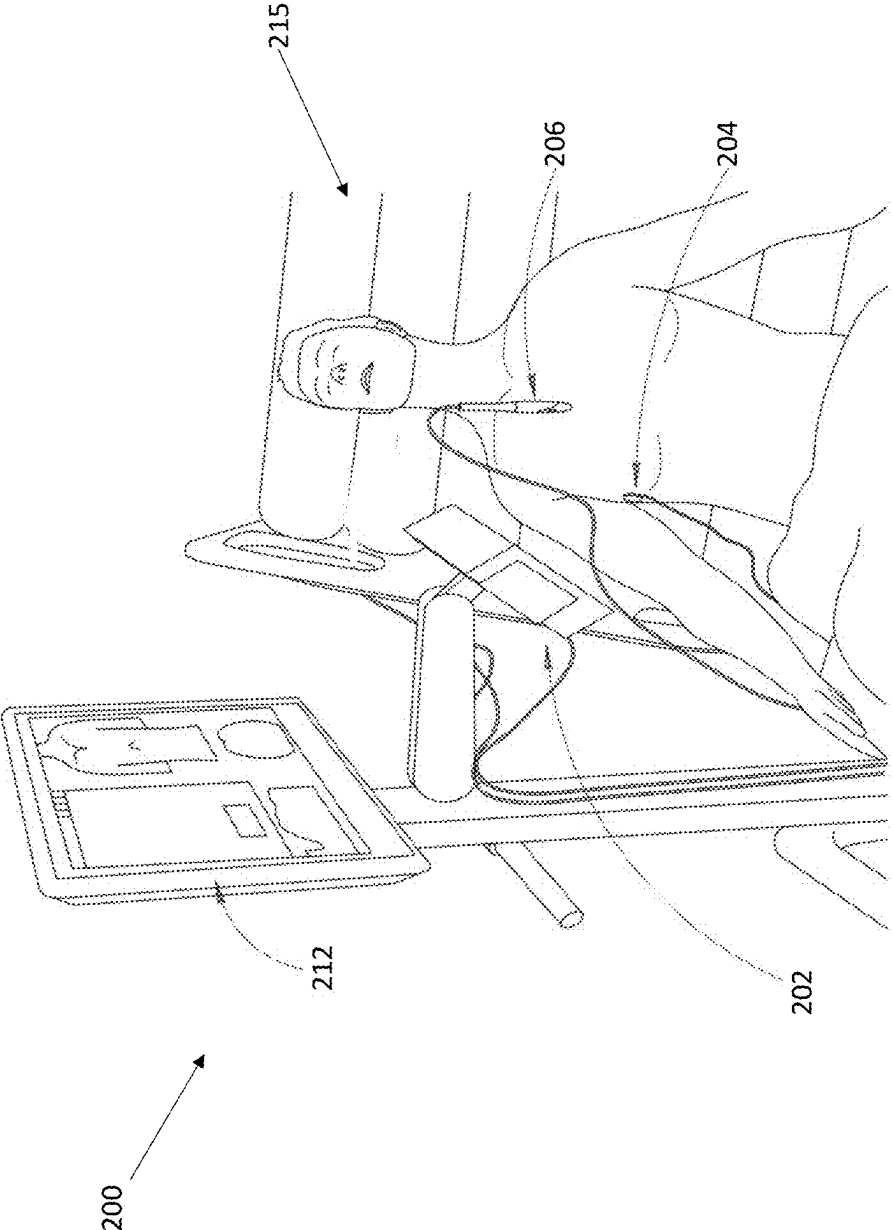
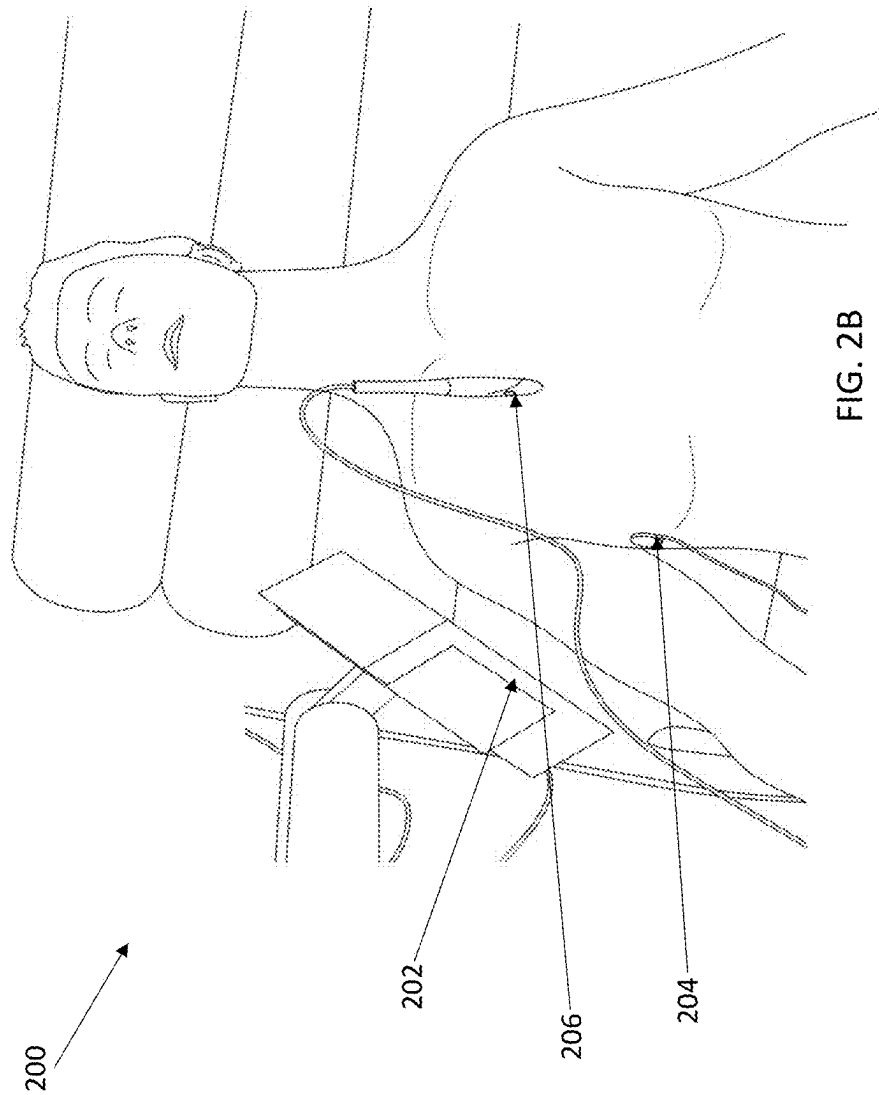
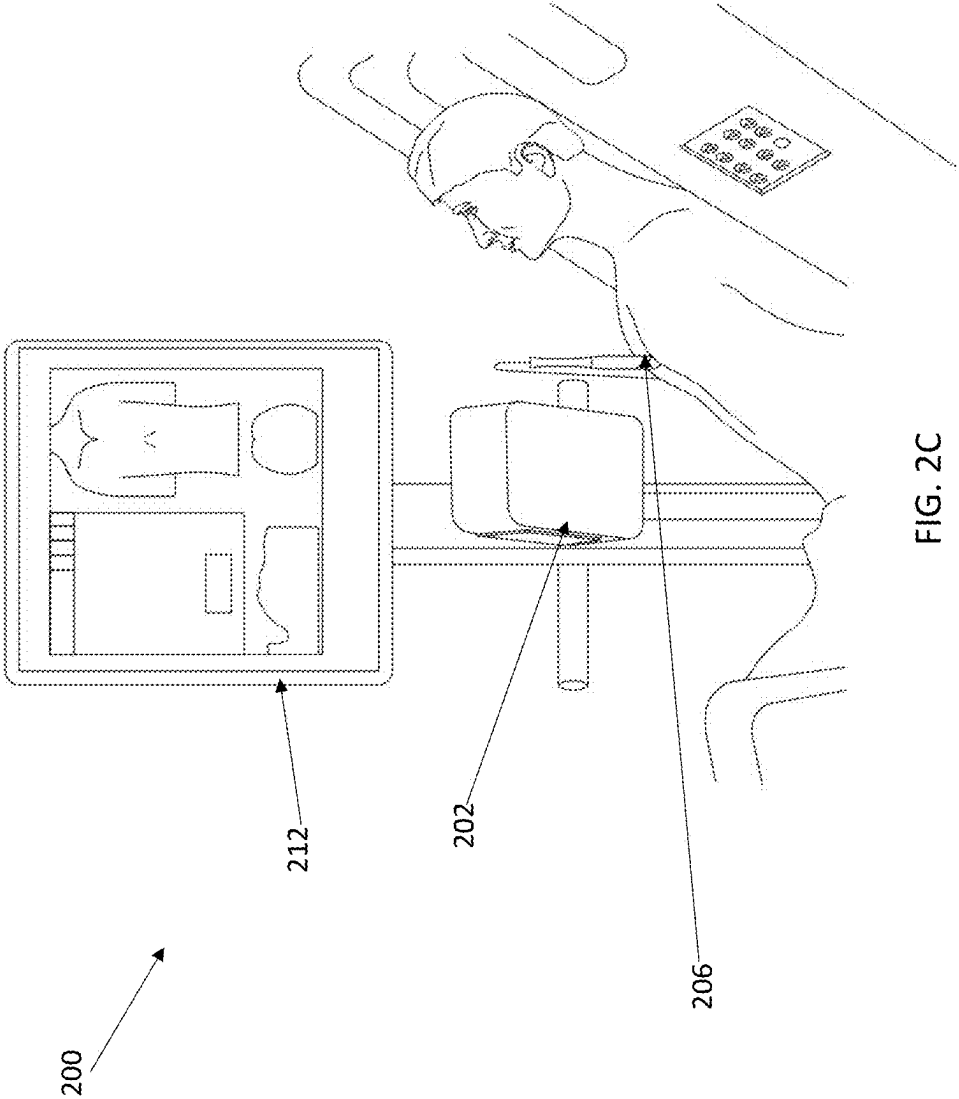


FIG. 2A





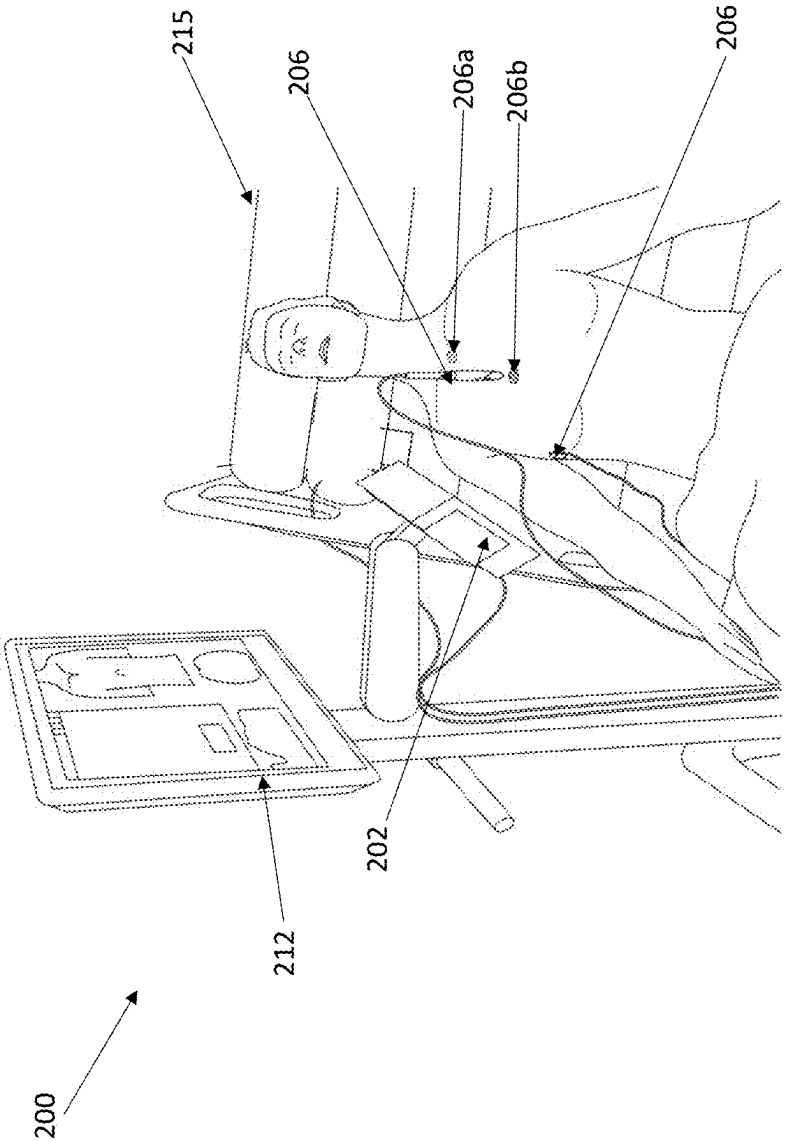


FIG. 2D

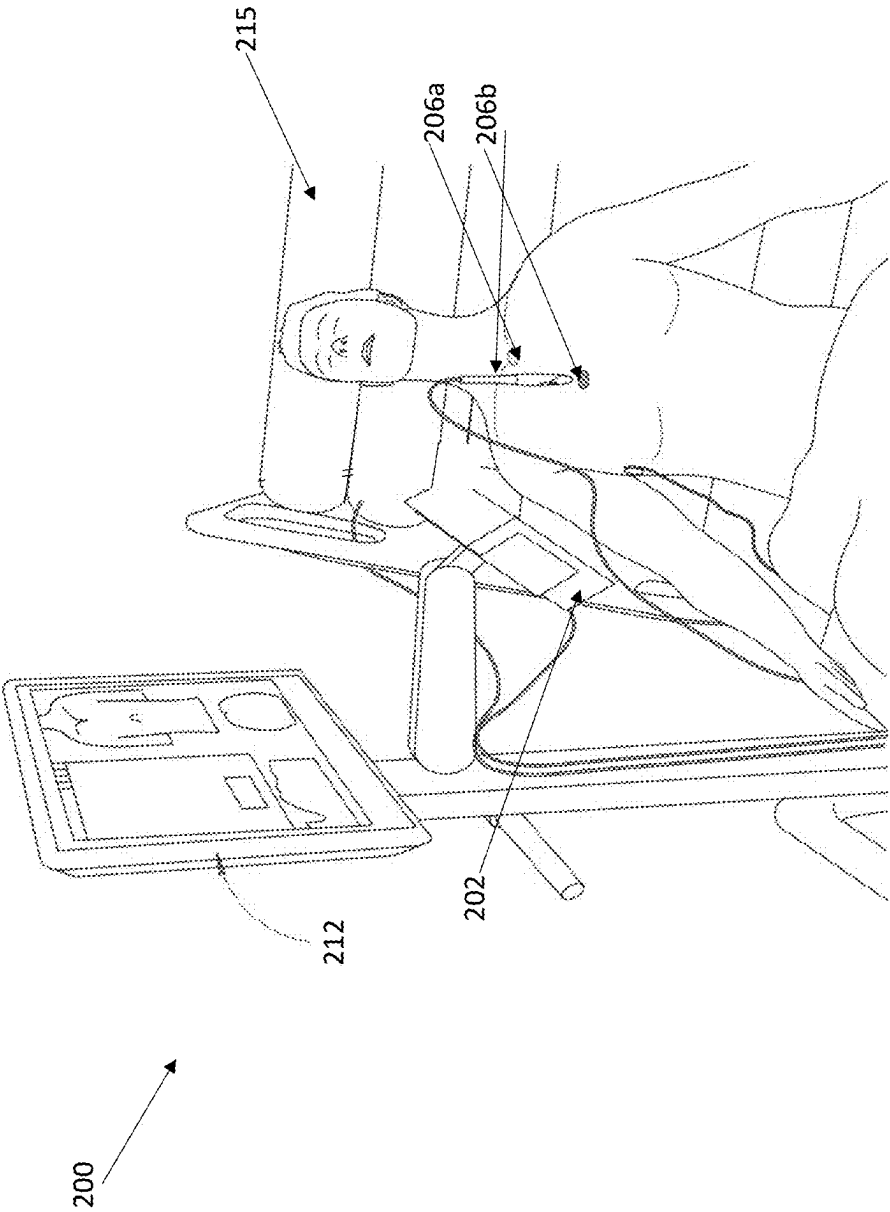


FIG. 2E

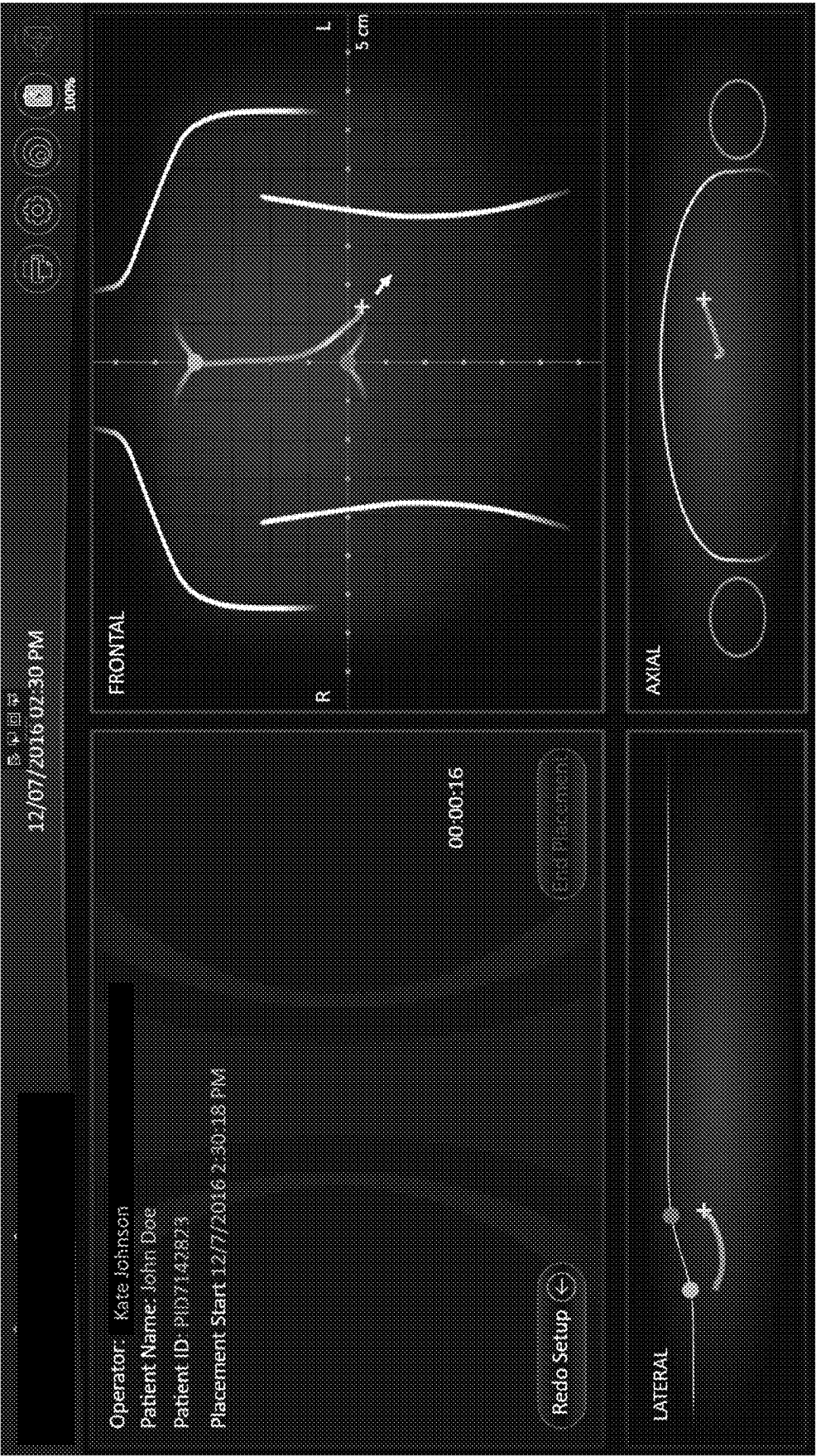


FIG. 3A

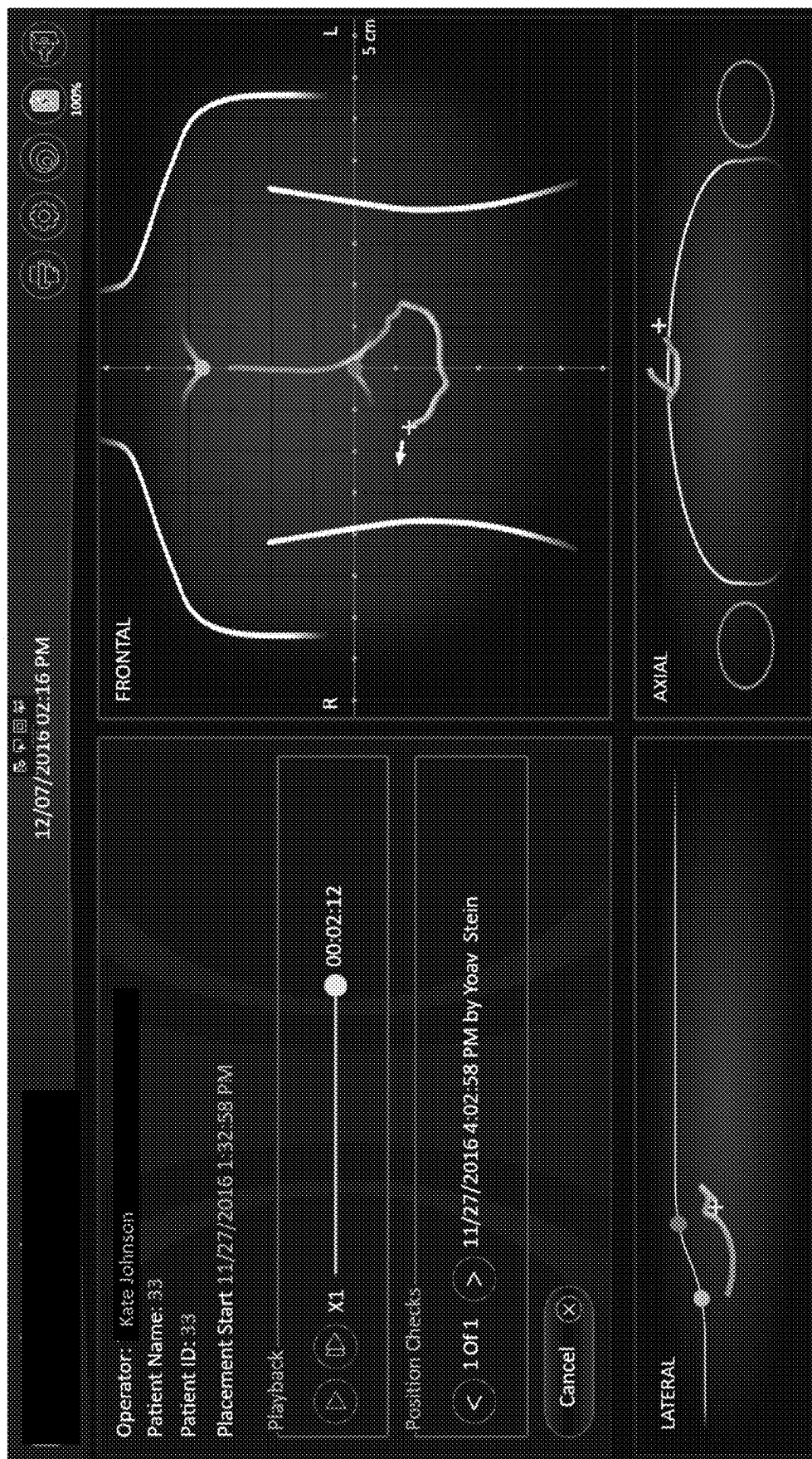


FIG. 3B

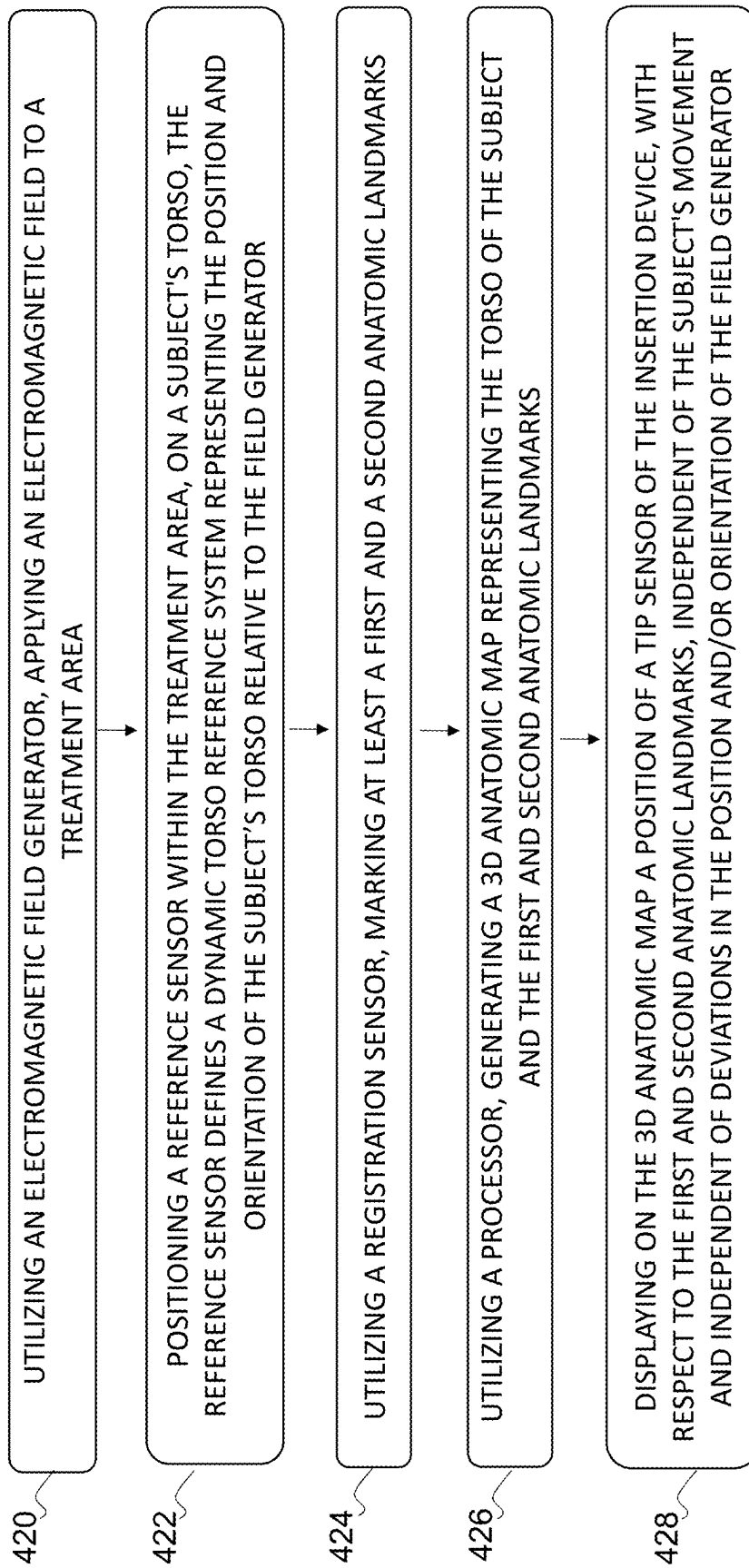


FIG. 4

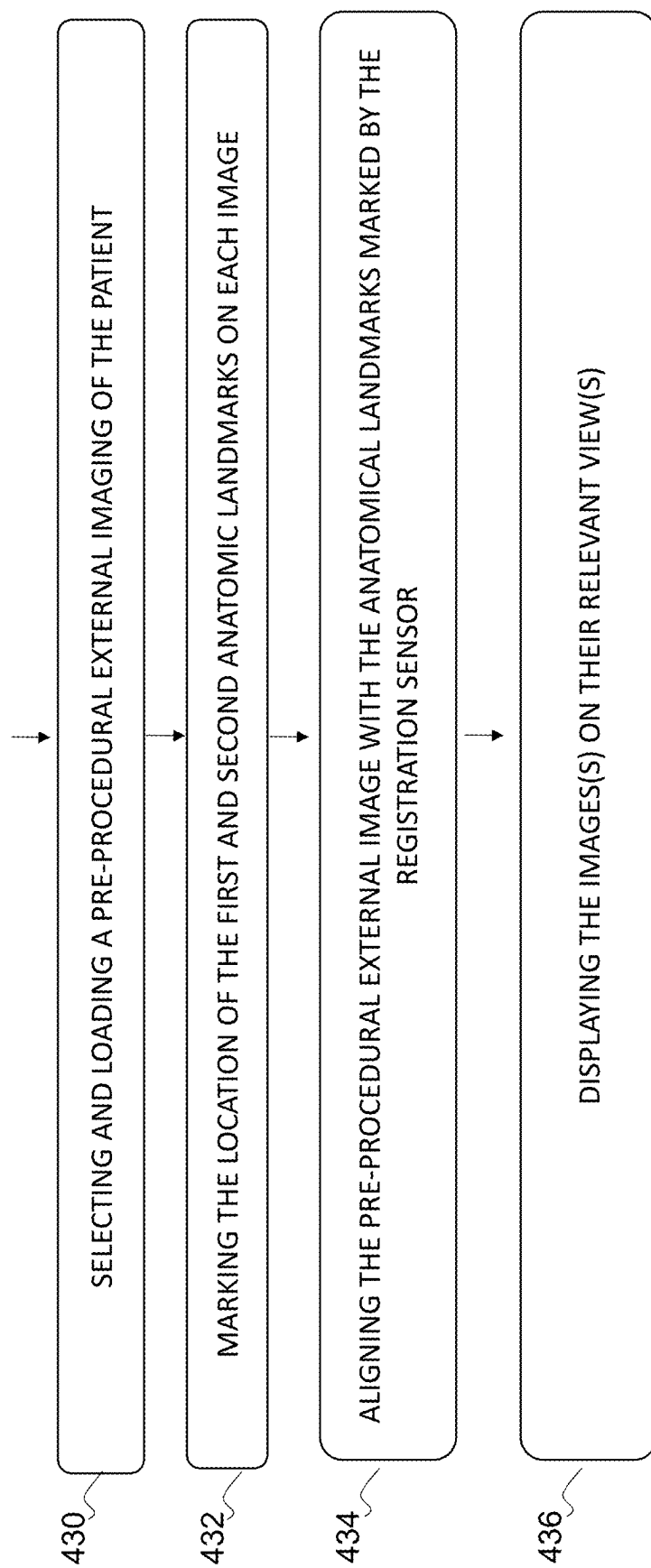


FIG. 4, cont.

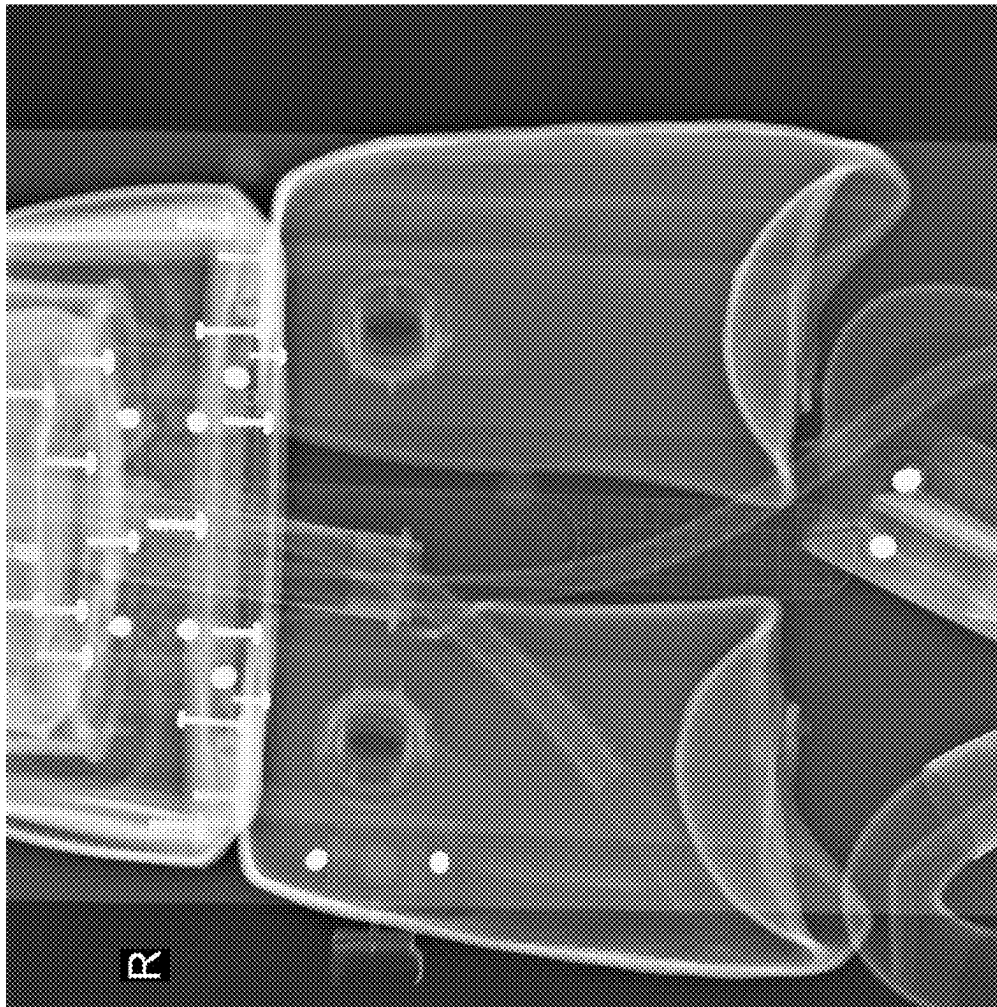


FIG. 5A

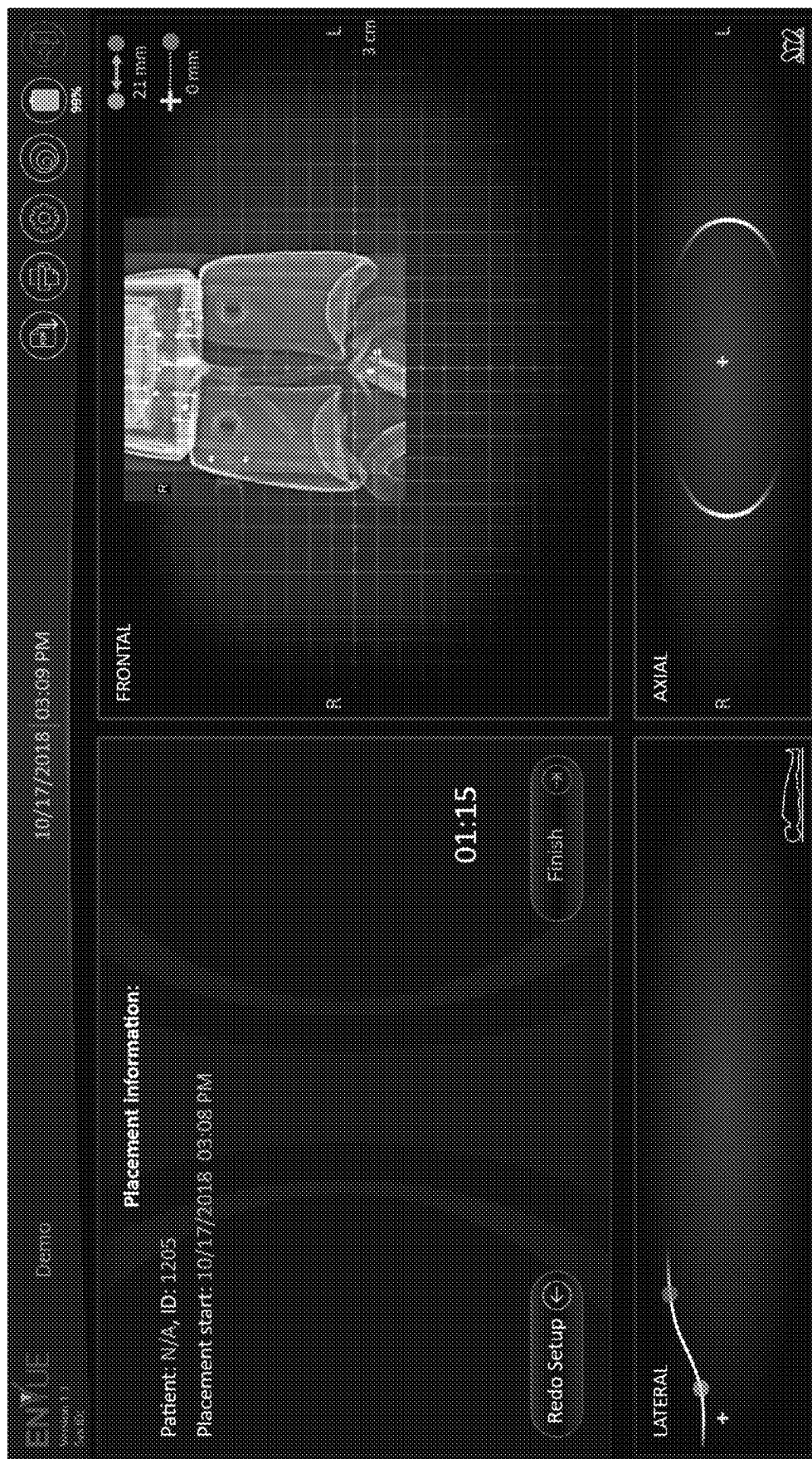


FIG. 5B

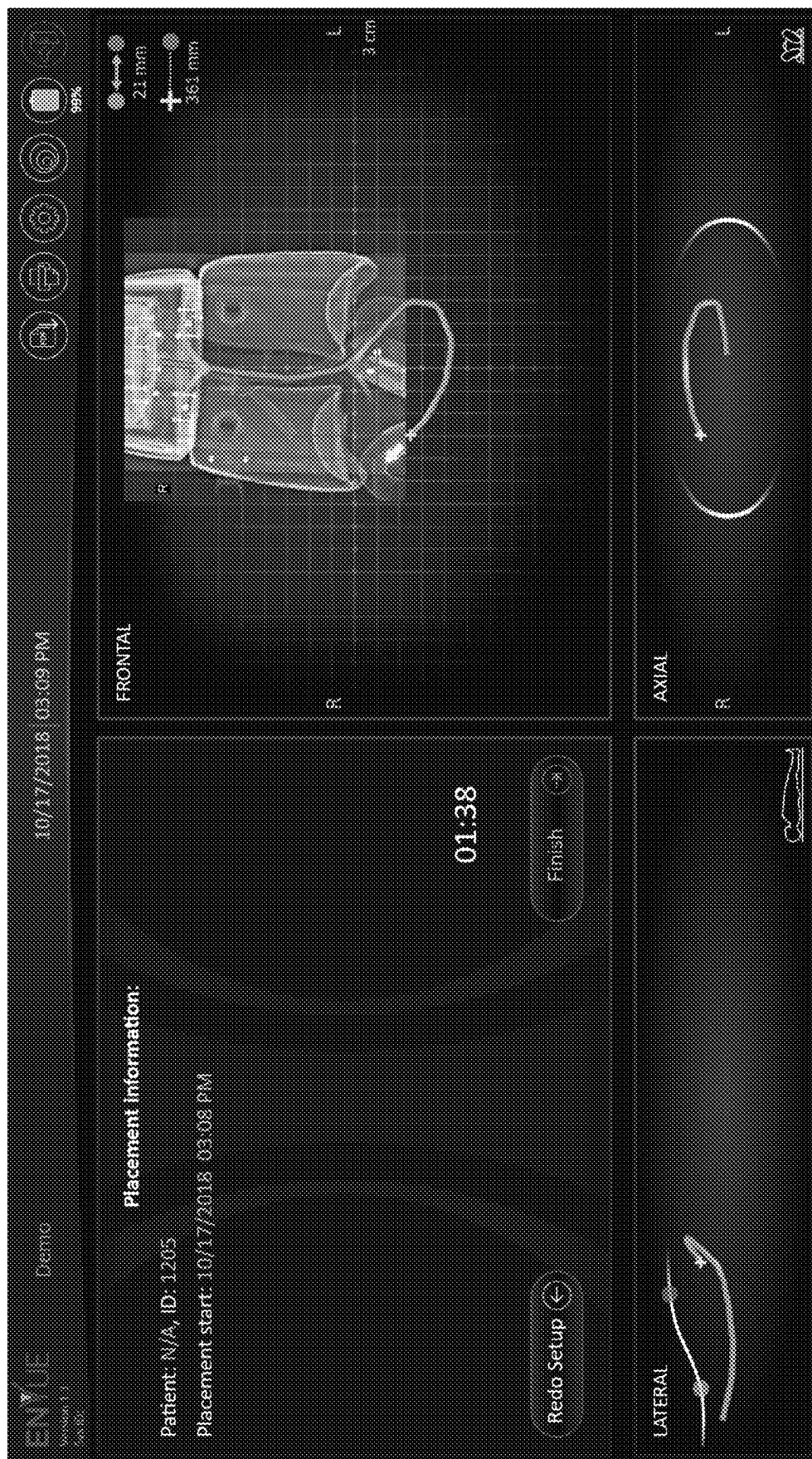


FIG. 5C

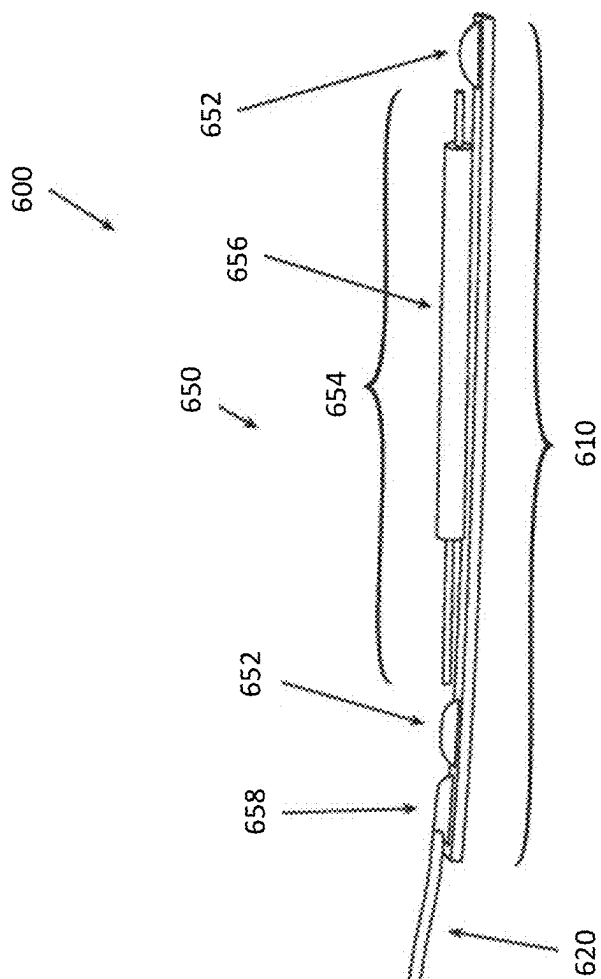


FIG. 6

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**INSERTION DEVICE POSITIONING
GUIDANCE SYSTEM AND METHOD****CROSS REFERENCE TO RELATED
APPLICATIONS**

This is a continuation of U.S. Pat. No. 17,561,620 filed Dec. 23, 2021 (issued as U.S. Pat. No. 11,382,701), which is a continuation in part of U.S. patent application Ser. No. 17/328,642 (issued as U.S. Pat. No. 11,779,403), filed May 24, 2021, which is a continuation of U.S. patent application Ser. No. 16/561,320 filed Sep. 5, 2019 (issued as U.S. Pat. No. 11,045,260), which claims priority to U.S. Provisional Patent Application No. 62/746,854 filed on Oct. 17, 2018. The contents of the applications are all incorporated herein by reference in their entirety.

FIELD OF INVENTION

Embodiments of the disclosure relate to insertion device positioning guidance systems and methods.

BACKGROUND

Enteral feeding is often used as nutritional support in patients unable to be fed otherwise. Although many benefits are associated with early initiation of enteral feeding, misplacement of feeding tubes is relatively common and can result in patient discomfort and complications. Confirming the position of the tube only after it is already inserted delays the feeding and the initiating of hydration or medication. Bedside electromagnetic (EM) systems for guided placement of enteral feeding tubes are available and are utilized by medical staff during the procedure to avoid misplacement of feeding tubes. There is still a need, however, for reliable real-time tracking systems that provide enhanced accuracy for critical tool positioning during medical procedures.

The foregoing examples of the related art and limitations related therewith are intended to be illustrative and not exclusive. Other limitations of the related art will become apparent to those of skill in the art upon a reading of the specification and a study of the figures.

SUMMARY

The following embodiments and aspects thereof are described and illustrated in conjunction with systems, tools and methods, which are meant to be exemplary and illustrative, not limiting in scope.

One of the problems often associated with an insertion of a feeding tube or catheter using an electromagnetic positioning guidance system, is that reliability is difficult to obtain in the subject environment, which is typically dynamic. For example, the subject often moves and the bed is moved from one place to another. There is thus provided herein an electromagnetic positioning guidance system reliably operable regardless of the subject's movement or position and which requires no calibration.

According to some embodiments there is provided a guidance system for positioning an insertion device comprising: an electromagnetic field generator configured to be positioned externally to a subject's torso, the electromagnetic field generator configured to generate an electromagnetic field covering a treatment area, an insertion device comprising an electromagnetic sensor, the electromagnetic sensor configured to receive signals indicative of the electromagnetic field, a display, and a processing circuitry

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configured to: load and display an X-ray, CT, ultrasound or MRI image of the subject's chest on the display, mark a location of a first and a second anatomic landmark on the subject's torso using a registration sensor and obtaining a subject coordinate system based thereon, identify the location of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image of the subject's chest; aligning the subject coordinate system with the loaded X-ray, CT, ultrasound or MRI image by aligning the registered first and second anatomic landmarks with the location of the first and second anatomic landmarks in the X-ray, CT, ultrasound or MRI image, and display, on the image, a path of the insertion device insertion with respect to the first and the second anatomic landmarks; wherein the path is generated according to changes in the strength of the electromagnetic field sensed by the tip sensor's during the insertion of the insertion device.

According to some embodiments, the identifying of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image is automatic and comprises applying image analysis algorithms and/or machine learning algorithms on the loaded X-ray, CT, ultrasound or MRI image.

According to some embodiments, the identifying of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image comprises marking the at least two anatomic landmarks using radiopaque markers, prior to the imaging.

According to some embodiments, the identifying of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image comprises positioning at least two markers on the subject's torso, prior to the imaging, and calculating the position of the anatomical landmarks based thereon.

According to some embodiments, the markers are configured to sense the electromagnetic field and to serve as reference sensors based thereon.

According to some embodiments, there is provided a guidance system for positioning an insertion device comprising: an electromagnetic field generator configured to be positioned externally to a subject's torso, said electromagnetic field generator configured to generate an electromagnetic field covering a treatment area; an insertion device comprising an electromagnetic sensor, said electromagnetic sensor configured to receive signals indicative of the electromagnetic field; a display; and a processing circuitry. According to some embodiments, the processing circuitry is configured to: load and display an X-ray, CT, ultrasound or MRI image of the subject's chest on the display, wherein the image is obtained after positioning of at least two radiopaque markers on predetermined positions on the subject's torso; mark the location of the at least two radiopaque markers on the patient's torso using a registration sensor; aligning the position of the radiopaque markers marked by the registration sensor with the position of the radiopaque markers in the image; and display, on the image, a path of the insertion device insertion with respect to the position of the radiopaque markers in the image; wherein the path is generated according to changes in the strength of the electromagnetic field sensed by the tip sensor's during the insertion of the insertion device.

According to some embodiments, the processing circuitry is further configured to load a predefined anatomical map representing a torso; and indicate the position of the first and second landmarks registered by the registration sensor on the map, wherein the aligning the registered first and second anatomic landmarks with the location of the first and second

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anatomic landmarks in the X-ray, CT, ultrasound or MRI image comprises aligning the predefined anatomical map with the X-ray, CT, ultrasound or MRI image.

According to some embodiments, the system further comprises the registration sensor configured to mark at least a first and a second anatomic landmarks on the subject's torso.

According to some embodiments, the registration sensor is incorporated into a tip of a stylus configured to be manually operated.

According to some embodiments, the first anatomic landmark is the suprasternal notch and the second anatomic landmark is the xiphoid process, and wherein a path display of the insertion device relative to the first and second anatomic landmarks is indicative of a successful insertion.

According to some embodiments, the system further comprises a reference sensor configured to define a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator.

According to some embodiments, the reference sensor is configured to be positioned, within the treatment area, on the subject's torso.

According to some embodiments, the reference sensor is configured to be positioned independently of the insertion of the insertion device.

According to some embodiments, the insertion device is a peripherally inserted central catheter (PICC).

According to some embodiments, the electromagnetic sensor is positioned at a distal tip of the insertion device.

According to some embodiments, the electromagnetic sensor is a separate unit configured for being removably positioned within the insertion device.

According to some embodiments, the insertion device comprises a sensor assembly, wherein the sensor assembly is removably positioned within a lumen of the insertion device, and wherein the sensor assembly comprises the electromagnetic sensor at a distal tip of the sensor assembly.

According to some embodiments, there is provided an insertion device positioning guidance system comprising: an electromagnetic field generator configured to generate an electromagnetic field covering a treatment area; a plate sensor configured to be positioned within the treatment area in a location defining an orientation of a subject (specifically, the vector perpendicular to the subject's chest); a reference sensor configured to be positioned, within the treatment area, on the subject's torso, the reference sensor is configured to define a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator; a registration sensor configured to mark at least a first and a second anatomic landmarks relative to the reference coordinate system; and processing circuitry configured to operate the field generator, read signals obtained from the plate sensor, the reference sensor and the registration sensor, calculate a position and orientation thereof relative to the field generator, the processing circuitry is further configured to generate an anatomic map, representing the torso of the subject and the first and second anatomic landmarks, the processing circuitry is further configured to facilitate visualization on the 3D anatomic map of a position, orientation and path of a tip sensor, located in a distal tip section of the insertion device, with respect to the first and second anatomic landmarks, independent of the subject's movement and independent of deviations in the position and/or orientation of the field generator, thus determination of a successful medical procedure is facilitated.

In some embodiments, the reference sensor is configured to be positioned on a side of the subject's torso, such that the

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3D anatomic map further depicts a body contour of the subject. In some embodiments, the 3D anatomic map shows a frontal upper view of the subject essentially parallel to the plate sensor. In some embodiments, the 3D anatomic map shows a side view of the subject essentially perpendicular to the plate sensor. In some embodiments, the 3D anatomic map shows an axial view of the subject.

In some embodiments, the system further comprises a monitor configured to display the 3D map.

According to some embodiments, the term "anatomic map" as disclosed herein, may refer to one or more schematic maps, one or more 2D anatomic maps, one or more 3D anatomic maps or any combination thereof. According to some embodiments, the term "anatomic map" as disclosed herein, may refer to a group of maps (e.g., 2, 3, 4 or more) maps, each representing a different view (for example, frontal view, frontal upper view, side view, axial view). According to some embodiments, the anatomic map or group of maps may refer to the subject chest image acquired prior to the procedure, X-Ray like images of the upper torso generated from CT or MRI scans taken prior to the procedure (for example, frontal view or side view DRRs—digitally reconstructed radiograph). This may advantageously allow the user to monitor the insertion of the insertion device (e.g. feeding tube, a catheter, a peripherally inserted central catheter (PICC)) on a scan of the subject showing for example the subject's lung and/or gastro-entestinal organs.

In some embodiments, the plate sensor is configured to be positioned under the subject's upper torso and/or neck.

In some embodiments, the registration sensor is a stylus configured to be manually operated.

In some embodiments, the insertion device is an enteral tube, such as but not limited to a gastroenteral or nasoenteral tube.

In some embodiments, the insertion device is a catheter. In some embodiments, the insertion device is a peripherally inserted central catheter (PICC).

In some embodiments, the first anatomic landmark is the suprasternal notch, and the second anatomic landmark is the xiphoid process. In some embodiments, a frontal view display of the path of the enteral tube relative to the first and second anatomic landmarks is indicative of a successful insertion. In some embodiments, the landmarks where the displayed path (in frontal view) of the enteral tube crosses an axis between the first and second anatomic landmarks is indicative of a successful insertion. In some embodiments, the actual shape of the displayed path of the enteral tube is indicative of a successful insertion.

According to some embodiments, there is provided a method for guiding an insertion device, the method comprising: utilizing an electromagnetic field generator, applying an electromagnetic field to a treatment area; positioning a plate sensor within the treatment area in a location defining an orientation of a subject; positioning a reference sensor within the treatment area, on a subject's torso, the reference sensor defines a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator marking at least a first and a second anatomic landmarks; utilizing processing circuitry, operating said field generator, reading signals obtained from the plate sensor, the reference sensor and the registration sensor, calculating a position and orientation thereof relative to said field generator, generating an anatomic map representing the torso of the subject and the first and second anatomic landmarks; and displaying on the anatomic map a position, orientation and/or path of a tip sensor of the insertion device,

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with respect to the first and second anatomic landmarks, independent of the subject's movement and independent of deviations in the position and/or orientation of said field generator.

According to some embodiments, marking at least a first and a second anatomic landmarks may include utilizing a registration sensor.

In some embodiments, the method further comprises the step of displaying the path of the distal tip section of the insertion device on the anatomic map and thus facilitates determination of a successful medical procedure.

In some embodiments, the positioning of the reference sensor comprises positioning thereof on a side of the subject's torso, such that the anatomic map further depicts a body contour of the subject.

In some embodiments, the positioning of the plate sensor comprises positioning thereof under the subject's upper torso and/or neck.

In some embodiments, the registration sensor is a manually operated stylus.

In some embodiments, the anatomic map shows a frontal upper view of the subject essentially parallel to the plate sensor. In some embodiments, the anatomic map shows a side view of the subject essentially perpendicular to the plate sensor. In some embodiments, the anatomic map shows an axial view of the subject.

In some embodiments, the insertion device is a enteral tube.

In some embodiments, the first anatomic landmark is the suprasternal notch and the second anatomic landmark is the xiphoid process. In some embodiments, a path display of the enteral tube relative to the first and second anatomic landmarks is indicative of a successful insertion. In some embodiments, the location where the displayed path of the enteral tube crosses an axis between the first and second anatomic landmarks is indicative of a successful insertion. In some embodiments, the actual shape of the displayed path of the enteral tube is indicative of a successful insertion.

In some embodiments, the electromagnetic field generator is not in any physical contact with the subject. In some embodiments, the electromagnetic field generator is designed not to be in physical contact with the subject. In some embodiments, the electromagnetic field generator is not designed to be in direct or indirect physical contact with the subject.

In some embodiments, the processor/processing circuitry may include two units or two sub-units. The first is configured to control the entire tracking system (e.g., operate the field generator, read signals obtained from the plate sensor, the reference sensor and the registration sensor and calculate a position and orientation thereof relative to said field generator. The second is configured to receive the calculated position and orientation information from the first processor, and use this information to generate a 3D anatomic map representing the torso of the subject and the first and second anatomic landmarks, and to allow visualization on the 3D anatomic map of a position, orientation and/or path of a tip sensor with respect to the first and second anatomic landmarks, independent of the subject's movement and independent of deviations in the position and/or orientation of the field generator.

According to some embodiments, the tip sensor, which may also be referred to as a position sensor, is positioned at or adjacent to the distal end of the tube/catheter/stylet and aids in determining the position of the distal end of the tube/catheter/stylet in a subject.

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The systems and methods described herein may be applied, according to some embodiments, using sensors (e.g., position sensors) such as magnetic field sensors, impedance-based sensors or ultrasonic sensors. According to some embodiments, the position sensor (e.g., tip sensor) may refer to an element mounted on a catheter/tube/stylet, which causes the processing circuitry to receive signals indicative of the coordinates of the element. The position sensor may include a receiver, which generates a position signal to a processing circuitry/control unit based on energy received by the sensor (for example, from the field generator). According to some embodiments, the communication between the sensors and the processing unit may be wireless.

More details and features of the current invention and its embodiments may be found in the description and the attached drawings.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, suitable methods and materials are described below. In case of conflict, the patent specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

BRIEF DESCRIPTION OF THE FIGURES

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

Exemplary embodiments are illustrated in referenced figures. Dimensions of components and features shown in the figures are generally chosen for convenience and clarity of presentation and are not necessarily shown to scale. The figures are listed below.

FIG. 1 is a block diagram of an insertion device positioning guidance system, in accordance with some embodiments;

FIG. 2A schematically illustrates an insertion device positioning guidance system in a hospital setting, in accordance with some embodiments;

FIG. 2B shows an enlarged portion of the illustration of FIG. 2A, in accordance with some embodiments;

FIG. 2C shows a side view of the illustration of FIG. 2A, in accordance with some embodiments;

FIGS. 2D-E schematically illustrate an insertion device positioning guidance system in a hospital setting, showing anatomic landmarks marked using a stylus, reference sensor and plate sensor located at different locations, in accordance with some embodiments;

FIG. 3A shows a view of a "live" display of placement of an insertion device, in accordance with some embodiments;

FIG. 3B shows a view of a "playback" display of placement of an insertion device, in accordance with some embodiments; and

FIG. 4 is a flow chart of the steps of a method for guiding the position of an insertion medical device, in accordance with some embodiments.

FIG. 5A shows an example of an upper body phantom simulation X-Ray used in some embodiments as the anatomic map;

FIG. 5B shows a view of a "live" display of placement prior to the insertion an insertion device, while using the

upper body phantom simulation X-Ray from FIG. 5A as the anatomic map, in accordance with some embodiments;

FIG. 5C shows a view of a “live” display of placement of an insertion device while using the upper body phantom simulation X-Ray from FIG. 5A as the anatomic map, in accordance with some embodiments; and

FIG. 6 shows a sensor assembly configured for incorporation into an insertion device, in accordance with some embodiments of the present invention.

DETAILED DESCRIPTION

Disclosed herein is a system and method for guiding insertion of an insertable medical device (e.g., a tube, such as a feeding tube). The disclosed system may be used as an insertion device positioning guidance system. The system may be used to track and indicate, in real time, the location of an insertion medical device during the insertion process. As one example, the system may track and indicate the location of a tip of a feeding tube as it is being inserted into the body of a subject. Advantageously, this makes the insertion procedure considerably easier and safer, ensuring that the tube is inserted at a correct location.

According to some embodiments, there is provided an insertion device positioning guidance system comprising: an electromagnetic field generator configured to generate an electromagnetic field covering a treatment area; a plate sensor configured to be positioned within the treatment area in a location defining an orientation of a subject (specifically, the vector perpendicular to the subject’s chest); a reference sensor configured to be positioned, within the treatment area, on the subject’s torso, the reference sensor is configured to define a reference coordinate system representing the position and orientation of the subject’s torso relative to said field generator; a registration sensor configured to mark at least a first and a second anatomic landmarks relative to the reference coordinate system; and processing circuitry configured to operate said field generator, read signals obtained from said the plate sensor, said reference sensor and said registration sensor, calculate a position and orientation thereof relative to said field generator, generate an anatomic map representing the torso of the subject and the first and second anatomic landmarks, said processor/processing circuitry is further configured to facilitate visualization on the anatomic map of a position, orientation and path of a tip sensor, located in a distal tip section of the insertion device, with respect to the first and second anatomic landmarks, independent of the subject’s movement and independent of deviations in the position and/or orientation of said field generator, thus determination of a successful medical procedure is facilitated. Optionally, the system further includes a monitor configured to display the map.

A plate sensor may be configured to be positioned within the treatment area in a location defining an orientation of a subject (specifically, the vector perpendicular to the subject’s chest). The orientation of a subject (specifically, the vector perpendicular to the subject’s chest) may be indicated by a plate sensor configured to be positioned within the treatment area. In a non-limiting example, the plate sensor is configured to be positioned under the subject’s upper torso and/or neck.

A reference coordinate system representing the position and orientation of the subject’s torso relative to the field generator may be indicated by a reference sensor configured to be positioned, within the treatment area, on a subject’s torso. The reference sensor may be positioned on a side of

the subject’s torso, such that the anatomic map further depicts a body contour of the subject.

The first and the second anatomic landmarks may be indicated by a registration sensor configured to mark at least the first and the second anatomic landmarks relative to the reference coordinate system. Optionally, the registration sensor is a stylus configured to be manually operated. Optionally, the first anatomic landmark is the suprasternal notch and the second anatomic landmark is the xiphoid process, and a path display of the enteral tube relative to the first and second anatomic landmarks is indicative of a successful insertion.

The electromagnetic field generator may be static throughout a duration of a procedure for placing a tube within a body of a subject. In such cases, a region covered by the electromagnetic field is static/constant throughout the duration of a procedure for placing a tube within a body of a subject. Advantageously, the static electromagnetic field may contribute to the accuracy of the display.

The anatomic map may show a frontal upper view of the subject essentially parallel to the plate sensor and/or a side view of the subject essentially perpendicular to the plate sensor and/or an axial view of the subject.

One example of hardware suitable for use as the above-mentioned electromagnetic tracking system, including the electromagnetic field generator and one or more of the sensors, is the Aurora® system by Northern Digital Inc., of Ontario, Canada.

Throughout the following description, similar elements of different embodiments of the device are referenced by element numbers differing by integer multiples of 100. For example, an electromagnetic field generator of FIG. 1 is referenced by the number 102, and an electromagnetic field generator of FIG. 2, which corresponds to electromagnetic field generator 102 of FIG. 1, is referenced by the number 202. According to some embodiments, the system may include a sensor assembly. According to some embodiments, the sensor assembly may be removably positioned within the insertion device. According to some embodiments, the sensor assembly may include a wire slidably positioned within the insertion device. According to some embodiments, the sensor assembly may include an electromagnetic sensor. According to some embodiments, the electromagnetic sensor may be positioned at a distal end of the wire, wherein the distal end of the wire may be positioned to correspond with a distal end of the insertion device. According to some embodiments, the wire may be configured to extend within (or through) the insertion device. According to some embodiments, the wire may be configured to extend within (or through) the insertion device such that the electromagnetic sensor is positioned at a distal end of the insertion device. According to some embodiments, the sensor assembly may be slidable within the insertion device. According to some embodiments, the sensor assembly may be less flexible than the insertion device, thereby guiding the position of the insertion device within the body of the patient during insertion thereof.

According to some embodiments, the electromagnetic sensor may be a passive electromagnetic sensor, which therefor enables the monitoring of the insertion device position and/or path, when subject to an electromagnetic field generator (such as, for example, an electromagnetic field generated externally to the patient’s body).

Advantageously, since the sensor is passive, (or in other words, does not transmit an electromagnetic field) a field generator external to the patient’s body is utilized. Accordingly, a larger electromagnetic field may be generated, which

is less sensitive to movements and therefore provides more reliable coordinates of the position of the insertion device. Such coordinates are critical for real-time monitoring of insertion device positioning including early detection of incorrect insertion, such as, for example, insertion into the patient's lungs instead of insertion to the stomach.

According to some embodiments, the sensor assembly may be removable from the insertion device, such that once the insertion device is positioned within the body of the patient, the sensor assembly may be removed therefrom without shifting the position of the insertion device.

The electromagnetic sensor may include a sensor body including a core. According to some embodiments, the core may be coupled to the wire configured to extend along at least a portion of the length of the insertion device. According to some embodiments, the core may be coupled to the wire configured to extend along the length of the insertion device. According to some embodiments, the electromagnetic core may be positioned at a distal end of the sensor assembly.

According to some embodiments, the sensor assembly may include a printed circuit board (PCB). According to some embodiments, the electromagnetic sensor may be coupled to the printed circuit board. According to some embodiments, the core and the wire may be directly or indirectly attached to the PCB. According to some embodiments, the PCB may be a FR-4 PCB.

According to some embodiments, the core may include a coil, such as a coil made of one or more copper wires wound around at least part of the core, also referred to herein as a "core assembly". According to some embodiments, the one or more copper wires may have a diameter of between 10 μ m and 70 μ m. According to some embodiments, the one or more copper wires may wound around the core between 40 and 3000 turns of wire around the core. According to some embodiments, the sensor body may have an outer diameter of 1 mm or less, such as but not limited to an outer diameter of 0.8 mm.

According to some embodiments, the ends of the one or more wires wound around the core may be soldered directly or indirectly (e.g. via a soldering coil) to a printed circuit board (PCB), such as but not limited to a FR-4 PCB. According to some embodiments, the PCB may be configured to process and/or signals produced by the core in response to an electromagnetic field to an external processing device and/or monitor via the wire running through the sensor lumen. According to some embodiment, the data generated by the processing circuit are indicative of a position of the sensor and thus of the tip of the insertion device.

According to some embodiments, the wire running along the sensor lumen may be a twisted wire, such as but not limited to a wire made of two intercalated and/or braided wires. According to some embodiments, the wire may be a pair of twisted copper wires. According to some embodiments, the wire may have an outer diameter of 0.5 mm or less, or 0.4 mm or less, such as but not limited to an outer diameter of 0.35 mm.

According to some embodiments, an RF induced heating of the insertion device in an MRI environment is below 5 degrees.

Advantageously, the insertion device, including the sensor assembly, as disclosed herein, exhibits a very low RF induced heating during MRI. Accordingly, the sensor assembly and/or the electromagnetic sensor may be formed as an integral part of the insertion device and does not need to be withdrawn for performing MRI procedures, to the convenience of both patients and caregivers.

This as opposed to other electromagnetic sensors/transmitters, which due to their RF induced heating must be taken out (either sensor or entire tube) prior to performing an MRI scan, in order to prevent internal damage being caused to the patient. This further obviates the need for reinsertion (if the position of the insertion device needs be verified), thereby enabling confirming the position of the insertion device without reintroducing the sensor, which re-introduction may be hazardous.

Reference is now made to FIG. 1 which is a block diagram of an insertion device positioning guidance system 100. System 100 includes an electromagnetic field generator 102 configured to generate an electromagnetic field 103a covering at least a region of interest 103b (e.g., a treatment area such as a subject's torso), a plurality of electromagnetic sensors, such as sensors 104, and 106, to indicate a position of a tip sensor (located in a distal tip section of the insertion device) on an anatomical map (FIGS. 3A-B) of the region of interest 103b (typically the subject's torso). System 100 further includes a processor 110 configured to operate said field generator, read signals obtained from the reference sensor and the registration sensor, calculate a position and orientation thereof relative to said field generator, and optionally generate an anatomic map representing the torso of the subject. Processor 110 is configured to facilitate visualization of the anatomic landmarks on an uploaded image (such as an X-ray, a CT scan, an ultrasound image or an MRI scan). System 100 further includes a monitor 112 operatively connected to processor 110 and configured to display, on the anatomic map and/or on the uploaded image, the positions of and/or the path leading from the insertion device tip to the insertion site. In some embodiments, monitor 112 may be integrated with processor 110, such as in the case of an all-in-one computer. A determination of a successful medical procedure (for example, an insertion of a feeding tube to the stomach as opposed to the lungs) is thus possible.

Sensor 104 is typically a reference sensor configured to be positioned on a subject's torso. Reference sensor 104 is configured to define a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator. Optionally, reference sensor 104 may be attached to the skin of the subject, for example on the side of a subject's torso such as beneath the subject's armpit. In such cases, the anatomic map further depicts a body contour of the subject. Reference sensor 104 may be, for example, a 6-DOF electromagnetic sensor, capable of determining 6 axes of its location (XYZ axes) and attitude (roll, yaw, and pitch) with respect to field generator 102.

Sensor 106 is typically a registration sensor configured to be positioned on and/or to mark at least a first and optionally also a second anatomic (thoracic) locations over the subject's body (e.g. the subject's torso). Different anatomical locations may be marked depending on the type of procedure used, the type of insertion medical device, etc. The marking of the anatomic location may be physical, such as attaching a marker/fiducial (such as a sticker). Alternatively, the marking of the anatomic location may be virtual, such as registering a virtual marker/fiducial. The marking, in accordance with embodiments, may facilitate identification or designation of an anatomical location within or on a subject's body such as, in a non-limiting example, a subject's suprasternal notch, and a subject's xiphoid process.

Optionally, registration sensor 106 is a stylus sensor configured to be manually operated to mark at least a first and a second anatomic location over the subject's body

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identified by the operator of the stylus. The marking may be made, merely as an example, by indicating to the software (for example, but not limited to, by pressing a GUI button or voice activation) once stylus sensor **106** is positioned over the desired point on the subject's body. The marking may be communicated to and registered by processor **110**.

According to some embodiments, the system may include a non-transitory computer-readable storage medium, or in other words, a memory module. According to some embodiments the memory module may have stored thereon one or more program codes configured to operate the processor and/or any one or more of the field generator **102**, sensors **104/106**, monitor **112**, and/or other elements of the system **100**.

According to some embodiments, the terms "program code", "software instructions", "software" and "algorithm" may be used interchangeably.

According to some embodiments, marking the anatomic locations may be automatic. According to some embodiments, marking the anatomic location may include utilizing one or more algorithms configured to search the image of the patient and then identify the anatomic locations thereon. According to some embodiments, marking the anatomic location may include utilizing one or more algorithms configured to automatically identify the anatomic locations.

According to some embodiments, processor **110** is configured to produce a subject coordinate system based on the signals obtained from reference sensor **104** and registration sensor **106**. According to some embodiments, processor **110** is further configured to align the subject coordinate system with the uploaded image by aligning the registered first and second anatomic landmarks with the location of the first and second anatomic landmarks in the uploaded image, and display, on the uploaded image, a path of the insertion device insertion with respect to the first and the second anatomic locations; wherein the path is generated according to changes in the strength of the electromagnetic field sensed by the tip sensor's during the insertion of the insertion device.

According to some embodiments, processor **110** is configured to automatically identify the first and the second anatomic locations on the uploaded image for example by applying image analysis algorithms and/or machine learning algorithms on the uploaded.

According to some embodiments, processor **110** is configured to identify the first and the second anatomic landmarks on the uploaded image based on radiopaque markers positioned at (or near) the least two anatomic landmarks on the subject's torso, prior to the imaging.

According to some embodiments, processor **110** is configured to identify the first and the second anatomic landmarks on the uploaded image based on radiopaque markers positioned on the subject's torso, prior to the imaging, and calculating the position of the anatomical landmarks, based on the position of the markers.

According to some embodiments, processor **110** is configured to identify the first and the second anatomic landmarks on the uploaded image based on radiopaque markers positioned on the subject's torso, prior to the imaging, and calculating the position of the anatomical landmarks, based on the position of the markers.

According to some embodiments, registration sensor **106** is further configured to register the position of the at least two markers positioned on the subject's torso. According to some embodiments, processor **110** is configured to align the

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subject coordinate system with the uploaded image by aligning the registered markers with the markers imaged in the uploaded image.

According to some embodiments, the anatomic locations may include any one or more of the suprasternal notch, the xiphoid process, a side of at least a portion of the torso of the patient, and a clavicle of the patient.

According to some embodiments, the registration may be done by touching, using the registration sensor, the radiopaque markers and aligning them with the position of the markers in the captured image.

System **100** is configured to work in conjunction with an insertion medical device (not shown), such as a feeding tube or a catheter, e.g., for example, a PICC. The insertion medical device may include one or more sensors to allow its tracking within region of interest **103b**. Preferably, the sensor is located at the tip of the insertion medical device. In such case, processor **110** and monitor **112** are configured to compute and display position and/or advancement of the tip of the insertion medical device between the designated anatomical locations leading to the insertion site/target area.

According to some embodiments, as used herein the terms "insertion device" and "insertion medical device" may refer to any device/tool adapted for insertion into a body. The insertion device may be any medical insertion device or a medical surgical device. Non-limiting examples of insertion medical devices include, feeding tubes, such as enteral tubes (for example, nasogastric feeding tubes), endotracheal tube, tracheostomy tube, stomach tube, catheter tubes or cricothyrotomy tube. Other examples of insertion devices are well known in the art.

According to some embodiments, the terms "processing circuitry" and "processor" may be used interchangeably.

In some embodiments, the insertion device is a tube. In some embodiments, the tube is a feeding tube. In some embodiments, the tube is a gastro/enteral feeding tube, such as, but not limited to, a nasogastric feeding tube or a naso-enteral feeding tube. According to some embodiments, the feeding tube may have disposed therein and/or thereon an electromagnetic sensor, for example at its distal end.

Reference is now made to FIGS. **2A-E** which schematically illustrate an insertion device positioning guidance system **200** in a hospital setting, in accordance with some embodiments. FIG. **2A** schematically illustrates an insertion device positioning guidance system in a hospital setting, in accordance with some embodiments, FIG. **2B** shows an enlarged portion of the illustration of FIG. **2A**, in accordance with some embodiments, FIG. **2C** shows a side view of the illustration of FIG. **2A**, in accordance with some embodiments and FIGS. **2D-E** schematically illustrate an insertion device positioning guidance system in a hospital setting, showing anatomic locations marked using a stylus, reference sensor (as shown in FIGS. **2A-C**) and plate sensor located at different locations, in accordance with some embodiments.

Similarly to system **100** of FIG. **1**, system **200** includes an electromagnetic field generator **202**, and a plurality of electromagnetic sensors **204** and/or **206**. Further, system **200** is configured to work in conjunction with an insertion medical device (not shown) which may include one or more electromagnetic sensors configured to sense and/or interfere with the electromagnetic field generated by field generator **202**. Optionally, monitor **212** of system **200** is integrated with a computer, which corresponds to or includes processor **110** of FIG. **1**.

According to some embodiments, electromagnetic field generator **202** may be positioned at such angle and position with respect to the subject, as to enable the generated

electromagnetic field to cover the external and internal working area, or in other words, the entire upper torso (at least from the nose area to the duodenum area). Reference sensor **204**, and stylus sensor **206** are all covered under the field produced by field generator **202**. The tip sensor of the feeding tube is configured to move inside the digestive system, and its path can thus be traced. Reference sensor **204** may be attached to and/or on the skin of the subject, for example beneath the subject's armpit. Suitable means for attachment of the sensor are well known in the art such as, for example, stickers, medical glue and the like. Reference sensor **204** may serve to detect location (XYZ axes) and attitude (roll, yaw, and pitch) of the subject with respect to field generator **202**, based on the electromagnetic field (not shown) emitted by field generator **202**.

Stylus sensor **206** may be manually operated to mark one or more anatomic locations over the subject's skin. For example, FIGS. 2D and 2E show the marking of two such anatomic locations (indicated as "**206a**" and "**206b**" in these figures) on the subject's chest. Anatomic location **206a** is marked over the suprasternal notch, and anatomic location **206b** is marked over the xiphoid process. The marking may be communicated to and registered by the computer.

Optionally, the computer receives signals of the locations and postures of reference sensor **204**, and the two marked anatomic locations **206a** and **206b**, and computes an anatomic mark representative of the subject's torso, thereafter the medical procedure can begin. In the exemplary case of guiding the insertion of a feeding tube, the tip of the feeding tube is equipped with a sensor. Optionally, the computer receives the actual position and orientation of the sensors from a second processor that receives the signals and calculates the sensors' locations. Optionally, the computer receives the actual position and orientation from a second processor that receives the signals from the sensors and calculates their physical location.

System **200** is operated as follows:

The electromagnetic field generator **202** is activated to apply an electromagnetic field to the treatment area, covering the subject's torso;

Reference sensor **204** is positioned within the treatment area, on a subject's torso, preferably on the side of the torso. Reference sensor **204** defines a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator;

Registration sensor **206** is used to mark two anatomic locations on the subject's torso (for example, the suprasternal notch and the xiphoid process);

Utilizing a processor, generating an anatomic map representing the torso and the two anatomic locations and displaying on monitor **212** the anatomic map and the position and path of the tip sensor (of the feeding tube). The path of the tip sensor may be displayed with respect to the two anatomic locations and/or with respect to a longitudinal axis passing between the two anatomic locations and along the center of the torso.

Reference is now made to FIG. 3A, which shows a view of a "live" display of placement of an insertion device, in accordance with some embodiments and to FIG. 3B, which shows a view of a "playback" display of placement of an insertion device, in accordance with some embodiments. Such displays may be presented on a monitor such as monitor **212**. The left corner includes general information and subject's details, and in the display of FIG. 3B, also playback controls.

The tip's path is schematically drawn, enabling the caregiver to visualize the entire insertion path of the tube, until

it reaches the desired location. Optionally, and as shown in FIGS. 3A and B, an arrow is near the tip of the path, indicating the actual direction to which the tube is pointing. Such arrow(s) may help the user to properly insert the tube (or better understand where and to which direction the tube is moving). The displays of both FIGS. 3A and B show three views of the subject's body: a frontal view shown at the top right side of the monitor, a lateral view shown at the bottom left side of the monitor, and an axial view shown at the bottom right side of the monitor. In some embodiments, different and/or additional views may be shown.

The caregiver inserting the insertion medical device can view the indications on monitor **212** while manually maneuvering the medical implement into the subject's body, so as to guide it to the desired location in the body. Reference is now made to FIG. 4, which is a flow chart of the steps of a method for guiding the position of an insertion medical device, in accordance with some embodiments. Step **420** includes application of an electromagnetic field to a treatment area. Optionally, an electromagnetic field generator, such as electromagnetic field generator **202** of FIG. 2, is positioned such that the electromagnetic field covers the treatment area. A reference sensor, such as reference sensor **204** of FIGS. 2A-D, is positioned on a subject within the treatment area, on a subject's torso, the reference sensor defines a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator (step **422**). Optionally, the reference sensor is positioned on a side of the subject's torso such as to indicate a body contour of the subject. At least a first and a second anatomic landmark are marked by utilizing a registration sensor (step **424**). Optionally, a stylus sensor, such as stylus sensor **206** of FIGS. 2A-D, is manually operated to mark the at least first and second anatomic locations (e.g., **206a** and **206b**). Alternatively, one or more registration sensors are positioned over the at least first and second anatomic locations. Optionally, the first anatomic location is the suprasternal notch and the second anatomic location is the xiphoid process. Each of reference sensor **204**, and the two marked anatomic locations **206a** and **206b** are present within the region of interest. Each of steps **420**, **422** and **424**, may be performed simultaneously or in an interchangeable order. An anatomic map representative of the subject's torso and the first and second anatomic locations is generated by utilizing a processor (step **426**). The anatomic map may be generated based on signals received from reference sensor **204** and two marked anatomic locations **206a** and **206b**. A position and orientation of a tip sensor of the insertion device is displayed on the anatomic map, with respect to the first and second anatomic locations, independent of the subject's movement and independent of deviations in the position and/or orientation of the field generator (step **428**). This can be accomplished when all sensors remain within the sensing volume of the field generator. Optionally, the method may further include selecting and loading a pre-procedural external imaging of the subject (e.g., chest X-Ray (such as the upper body phantom simulation X-Ray shown in FIG. 5A), CT, ultrasound or MRI) (step **430**), marking (automatically or manually by the user) on the loaded Frontal image (in case of chest C-Ray) or Frontal and Lateral images (generated DRRs from CT or MM), the first and second anatomic landmarks (step **432**), and overlaying the image (X-Ray) or images (Frontal and Lateral), while aligning the marked anatomic locations to their predefined locations (step **434**). The pre-procedural external image is aligned with the reference coordinate system. According to some embodiments, the alignment may be accomplished by identifying the first

and the second anatomic landmarks in the pre-procedural external image by applying image analysis algorithms and/or machine learning algorithms on the image, and aligning the identified landmarks with the registered landmarks. Alternatively, the alignment may be accomplished by positioning radiopaque markers at the anatomical landmarks prior to the imaging and aligning the position of the markers visualized in the image with the registered landmarks. Yet alternatively, the alignment may be accomplished by positioning radiopaque markers at known positions on the subject's torso prior to the imaging, calculating the location of the anatomical landmarks based on the position of the radiopaque markers and aligning the calculated location of the anatomical landmarks with the registered landmarks. Yet alternatively, the alignment may be accomplished by positioning radiopaque markers on the subject's torso, registering the position of the markers in the reference coordinate system by touching the markers with registration sensor and aligning the position of the registered markers with the markers imaged in the pre-procedural external image. In step 436, the path of insertion may be displayed on the pre-procedural external image based on the alignment.

Optionally, the anatomic map enables visualization of the location of an insertion device (having an electromagnetic sensor configured to sense and/or interfere with the electromagnetic field generated by the field generator) within a subject's body by computing and displaying a position of the inserted device vis-à-vis first and second anatomic locations 206a and 206b. Optionally, the computing may include normalizing the position based at least on signals received from reference sensor 204 and obtained continuously and/or in real-time during the procedure.

Reference is now made to FIG. 5B which shows a view of a "live" display of placement prior to insertion an insertion device, while using the chest-simulation X-Ray of FIG. 5A as the anatomic map, in accordance with some embodiments. It is understood that other types of imaging may likewise be used and is thus within the scope of the disclosure. Following the upload of the X-Ray, the user marks on the loaded Frontal image (or Frontal and Lateral images when DRRs from CT or MRI are used), the first and second anatomic locations thus enabling an overlaying of the X-Ray image, while aligning the marked anatomic locations to their predefined locations on their respective views.

Once the alignment is completed, the insertion of the insertion device (e.g. feeding tube, catheter, and/or PICC) may be monitored on the actual X-Ray image, as shown in FIG. 5C, thus providing a subject specific view of the insertion process.

Reference is made to FIG. 6, which shows a sensor assembly 600 configured for incorporation into an insertion device, in accordance with some embodiments of the present invention. According to some embodiments, the electromagnetic assembly 600 includes a PCB 610, to which a sensor body 650 is coupled, such as, for example, via a soldering coil 652. According to some embodiments, the sensor body 550 includes a core 654 wrapped around which is a copper coil 556. According to some embodiments, the PCB 650 may be configured to process and/or transmit signals, produced by core 656 in response to an electromagnetic field, to an external processing device and/or monitor (not shown) via a wire 620 soldered or otherwise connected to PCB 650. According to some embodiment, the data generated by PCB 650 are indicative of a position of sensor assembly 600 and thus of the tip of the insertion device, within a patient's body. According to some embodiments, the wire 620 may be

a twisted wire, made of two intercalated/braided wires, which advantageously was found to cause an RF induced heating (ΔT) of below 2 degrees in an MRI environment using a 64 MHz RF coil. However, it is understood that other wires configured to have an RF induced heating (ΔT) of below 5, 4, 3 or 2 degrees in an MRI environment using a 64 MHz RF coil, may likewise be utilized. According to some embodiments, the sensor body 650 has an outer diameter of less than 1 mm and the wire 620 an outer diameter of less than 0.4 mm making them suitable for incorporation into an insertion device without causing a significant increase in the outer diameter of the insertion device.

Advantageously, by incorporating sensor assembly 600 into an insertion device, the field generator applied (not shown) may be external to the patient, thus enabling generating a larger field which is less sensitive to movement of the patient and thus of the sensor relative to the field generator. In addition, by having sensor assembly 600 positioned within the insertion device, re-confirmation and/or readjustment of tube position may be performed without reintroducing a stylet, which reintroducing may cause undesired movement of the insertion device within the patient as well as cause physical harm during the procedure.

According to some embodiments, the electromagnetic assembly is configured to move inside the digestive system of the patient with the movement of the insertion device. According to some embodiments, the electromagnetic assembly may be configured to guide the movement of the insertion device within the digestive system of the patient. Advantageously, the sensor assembly may be configured to trace the path of the sensor assembly and/or the insertion device during the positioning thereof in the body of the patient.

Advantageously, by having sensor assembly 600 removably positioned within the insertion device, once the insertion device is positioned at a desired position within the body of the patient, the sensor assembly 600 may be removed therefrom.

The present invention may be a system, a method, and/or a computer program product. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the present invention.

The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light

pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire. Rather, the computer readable storage medium is a non-transient (i.e., not-volatile) medium.

Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

Computer readable program instructions for carrying out operations of the present invention may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like, and conventional procedural programming languages, such as the "C" programming language or similar programming languages. The computer readable program instructions may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the present invention.

Aspects of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer readable program instructions.

These computer readable program instructions may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions

stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

1. A processing circuitry configured to:

load and display an X-ray, CT, ultrasound or MRI image of a subject's chest on the display;

receive from a registration sensor a location of a first and a second anatomic landmark on a subject's torso using a registration sensor,

identify the location of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image of the subject's chest; and

align the subject coordinate system with the loaded X-ray, CT, ultrasound or MRI image by aligning the registered first and second anatomic landmarks with the location of the first and second anatomic landmarks in the X-ray, CT, ultrasound or MRI image.

2. The processing circuitry of claim 1, further configured to receive signals indicative of a strength of an electromagnetic field sensed by a tip sensor positioned on a distal end of an insertion tube.

3. The processing circuitry of claim 2, wherein the electromagnetic field is generated by a field generated positioned

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externally to the subject's chest and wherein the electromagnetic field covers the subject's chest.

4. The processing circuitry of claim 2, wherein the signals indicative of the strength of the electromagnetic field are obtained continuously during insertion of the insertion device. 5

5. The processing circuitry of claim 3, further configured display, on the aligned X-ray, CT, ultrasound or MRI image, a path of the insertion device during its insertion; wherein the path is generated based on the aligned coordinate system, the first and the second anatomic locations and changes in 10 the strength of the electromagnetic field sensed by the tip sensor's during the insertion.

6. The processing circuitry of claim 1, wherein the identifying of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image is automatic and comprises applying image analysis algorithms on the loaded X-ray, CT, ultrasound or MM image. 15

7. The processing circuitry of claim 1, wherein the identifying of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image comprises marking the at least two anatomic locations using radiopaque markers, prior to the imaging. 20

8. The processing circuitry of claim 7, wherein the markers are configured to sense the electromagnetic field and to serve as reference sensors based thereon.

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9. The processing circuitry of claim 1, wherein the identifying of the first and the second anatomic landmarks on the loaded X-ray, CT, ultrasound or MRI image comprises positioning at least two markers on the subject's torso prior to the imaging and calculating the position of the anatomical landmarks based thereon.

10. The processing circuitry of claim 1, wherein said processing circuitry is further configured to load a pre-defined anatomical map representing a torso; and to align the map, prior to the loading of the image of the subject's chest.

11. The processing circuitry of claim 1, wherein the first anatomic location is the suprasternal notch and the second anatomic location is the xiphoid process.

12. The processing circuitry of claim 1, wherein the insertion device is a peripherally inserted central catheter (PICC).

13. The processing circuitry of claim 1, wherein the insertion device is a feeding tube.

14. The processing circuitry of claim 1, wherein the electromagnetic sensor is positioned at a distal tip of said insertion device.

15. The processing circuitry of claim 1, wherein the electromagnetic sensor is a separate unit configured for being removably positioned within the insertion device.

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